

Native Budget Dual Certificates: Exact Ridge–KKT Reciprocity and a Rational Supporting-Hyperplane Atlas

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Abstract

The preceding finite native-profile study measured the source norm required to approximate signed prime-shell targets using literal Möbius cutoff profiles. We derive a target-space dual certificate for that budget frontier. If M is the source Gram matrix and V the physical image map, the ridge system $(V^T V + \rho M)c_\rho = V^T b$ yields, for every $\rho > 0$, the lower bound

$$D_\rho = \frac{\|b\|^2 - R^2 - b^T V c_\rho}{\rho} \leq \min_{\|Vc - b\| \leq R} c^T M c.$$

At an active finite frontier this is equality. We also isolate the reciprocal convention $\mu = 1/\rho$ between the KKT multiplier and the ridge parameter, correcting a notation ambiguity in the preceding numerical release without changing its values. On the frozen 18-row grid, 72 twenty-digit rational ridge choices produce exact rational dual fractions; all are within the stated 0.999999999 relative tightness floor. The result is a finite structural certificate, not an asymptotic estimate.

1 Question and finite setting

The TPC-299 release [1] converted a finite profile-angle ladder into a native source-budget frontier. For a literal prefix U_k , a frozen physical shell matrix A , and

$$V_k = A^T U_k, \quad M_k = U_k^T U_k,$$

the normalized residual target is represented by

$$B_R(b) = \min\{c^T M_k c : \|V_k c - b\|_2 \leq R\}. \quad (1)$$

Here $R = \frac{1}{2}\sqrt{|S|}$ on a shell S . The question in this paper is not whether (1) can be solved numerically, but whether its obstruction can be exported as a checkable target-space witness. Such a witness is useful because it separates the finite convex calculation from the choice of a primal optimizer.

All matrices in the declared fixture are rational. The source profiles are

$$\beta_z(t) = \lambda(t) - \sum_{\substack{d \leq z \\ d|t}} \mu(d), \quad z \in \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61\}. \quad (2)$$

The finite audit has 18 rows and 1,380 shell edges. It is deliberately restricted: no moving-shell limit or arithmetic L^2 statement is inferred.

2 Dual formula

Theorem 1 (weak native-budget dual). *Let $M \succ 0$, let V and b be finite real data, let $R \geq 0$, and let $\rho > 0$. Define*

$$(V^T V + \rho M)c_\rho = V^T b. \quad (3)$$

Then

$$D_\rho = \frac{\|b\|_2^2 - R^2 - b^T V c_\rho}{\rho} \quad (4)$$

is a lower bound for the value in (1), whenever (1) is feasible.

Proof. Set $\mu = 1/\rho$ and consider

$$L(c, \mu) = c^T M c + \mu(\|Vc - b\|_2^2 - R^2).$$

Its quadratic part is positive definite. Its unique minimizer satisfies

$$Mc + \mu V^T (Vc - b) = 0,$$

which is exactly (3). Multiplying (3) by c_ρ^T gives

$$\inf_c L(c, \mu) = \frac{\|b\|_2^2 - R^2 - b^T V c_\rho}{\rho}.$$

For any feasible c , $L(c, \mu) \leq c^T M c$; taking the infimum and then the primal minimum proves (4). $\square$

The formula is a supporting-hyperplane certificate in target space. Indeed, after whitening with $y = M^{1/2}c$ and $W = VM^{-1/2}$, the residual of the ridge point is

$$e_\rho = W y_\rho - b = -\rho(WW^T + \rho I)^{-1}b. \quad (5)$$

The normal vector $-e_\rho$ supports the radius- R target ball when $\|e_\rho\| = R$. Thus the dual scalar is not an extra fitted statistic: it is the value of a target-space supporting functional.

Proposition 1 (active strong duality). *If*

$$\text{dist}(b, \text{range } V) < R < \|b\|_2$$

and b has a visible component in $\text{range } V$, there is a unique $\rho_ > 0$ with $\|Vc_{\rho_*} - b\|_2 = R$, and*

$$D_{\rho_*} = B_R(b) = c_{\rho_*}^T M c_{\rho_*}. \quad (6)$$

Proof. In singular coordinates of W , the squared residual is

$$\|b_\perp\|_2^2 + \sum_j \left(\frac{\rho}{\sigma_j^2 + \rho} \right)^2 a_j^2.$$

It is continuous, begins at the least-squares residual, and increases to $\|b\|_2^2$. The stated hypotheses give a unique crossing. At the crossing, $\mu = 1/\rho_*$ satisfies the convex KKT conditions; Slater duality gives (6). $\square$

Table 1: TPC-300 finite rational dual atlas.

quantity	certified count/value
rows / shell edges	18 / 1,380
exact rational dual cases	72/72
dual/primal ratio lower bound	> 0.999999999 in 72/72
weighted threshold ratio $> 9 \cdot 10^{-5}$	18/18
weighted threshold ratio $> 5 \cdot 10^{-4}$	15/18
weighted threshold ratio $> 10^{-3}$	14/18
weighted full-prefix ratio $> 10^{-3}$	11/18

3 The reciprocal parameter correction

There are two positive parameters in common presentations of this path. The ridge parameter ρ appears in (3), while the KKT multiplier μ multiplies the squared residual constraint. Stationarity gives

$$\rho = \frac{1}{\mu}. \quad (7)$$

The TPC-299 producer used the matrix in (3), so its stored field `lagrange_multiplier` is numerically a ridge parameter. TPC-300 records this as a scoped notation erratum. None of the source budgets or finite obstruction counts changes. The distinction matters when comparing dual values: substituting ρ where μ is required changes the stationarity equation and generally misses the active residual.

4 Exact rational certificate compiler

For rational V, M, b, R^2 and rational $\rho > 0$, equation (3) is a rational linear system. Exact Gaussian elimination over $\mathbb{Q}$ produces $c_\rho \in \mathbb{Q}^k$, and (4) is an exact rational number. The compiler therefore stores two layers:

1. the numerator and denominator of the selected rational ρ ;
2. decimal enclosures for the dual budget and hashes of the exact dual fraction and coefficient vector.

The selected ρ is the 20-significant-digit rational midpoint of the frozen TPC-299 ridge interval. This is not claimed to be the exact irrational optimizer; it is a weak-dual point whose exact fraction can be replayed without a floating-point optimizer. Its tightness is measured against the parent certificate’s outward presentation enclosure.

5 Finite audit

The audit replays three threshold targets (weighted minimum, unit-edge max-cut, and all-positive) and the full available prefix for the weighted minimum on every row. This gives $18(3+1) = 72$ cases. Table 1 summarizes the claim-bearing counts.

The smallest recorded dual-to-parent-primal lower bound is 0.999999999999999962310666478. The exact-fraction hashes and coefficient hashes are recomputed by a producer-independent source-first checker in normal and optimized Python modes. Scalar and two-coordinate exact stress fixtures separately verify weak duality, equality at an active point, and the reciprocal correction.

6 What this does and does not close

The strongest positive result is a reusable conversion

native profile prefix $\longrightarrow$ ridge system $\longrightarrow$ exact rational target-space dual witness.

It turns the TPC-299 budget obstruction into a lower-bound certificate that does not trust the primal source coefficients as its final evidence. The strongest obstruction is unchanged but independently transported: on 14 of 18 threshold rows, and 11 of 18 full-prefix rows, the weighted target still requires more than 10^{-3} of the frozen source norm squared.

The finite profile family, shell grid, target tolerance, and normalization remain modeling choices. No growing profile-budget theorem, arithmetic L^2 estimate, fixed-power credit, full Gate B closure, or twin-prime result is claimed. The natural next test is hostile variation of tolerance and source normalization, so that the dual gap cannot be attributed to one finite convention.

Reproducibility and claim status

The project contains the exact-fraction producer, an independent replay, an exact stress suite, a canonical JSON certificate, and a Bridge-B checker. The Session-named Route-A/Route-B evaluator files are absent from the checkout; consequently this paper reports local fail-closed validation only. The status is

PROVED_EXACT_FINITE_NATIVE_BUDGET_DUALITY_AND_RECIPROCAL_
MULTIPLIER_CORRECTION_PLUS_NUMERICALLY_CERTIFIED_FINITE_
RATIONAL_DUAL_WITNESS_ATLAS.

References

- [1] Liang Wang. Native profile budget frontier, 2026. TPC-299 project release in the accompanying repository.